

SOLUTION ARCHITECTURE

Date: 18th July 2026	Project Name: Book a Doctor App	Team Lead: Kilari Ushodaya
Domain: Healthcare	Team Members: Shyam Prasad Mantri, Madhu Hasitha Kondeti, Nanda Kumar Madugula, Mantena Sirisha	

3-Tier Layered Architectural Structure

Layer Name	Component Role in Architecture	Technologies Used
Presentation Layer	Renders responsive user UI for doctor search, slot booking, file uploads, and dashboards.	HTML5, Vanilla CSS3 (Glassmorphism), JS ES6+
API Gateway Layer	Handles incoming REST requests, enforces security headers, and routes endpoints.	Flask Blueprint Routes / WSGI
Core Logic Services	Auth Svc, Triage Engine, Booking Engine, Document Vault, Notification Gateway.	Python 3.10+
Data Storage Layer	Persists operational appointment records and encrypted diagnostic files.	SQLite / PostgreSQL, Local Vault